

ECE 210 Midterm 3 Worksheet

Note: This worksheet is not guaranteed to be entirely representative of the midterm's contents. Material may appear on this worksheet which will not appear on the midterm, and vice versa.

Partial Fraction Expansion

1) (Technique 1)

a) sub problem 1

b) sub problem 2

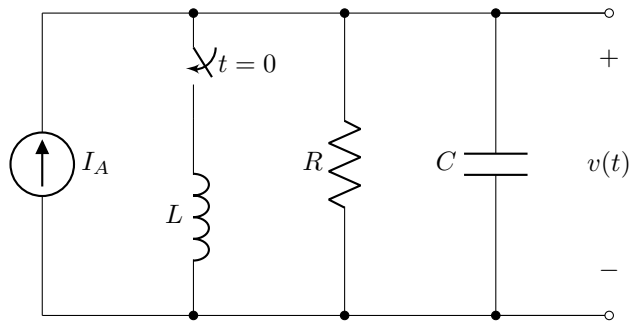
2) (Technique 2)

a) sub problem 1

b) sub problem 2

S-Domain Circuit Analysis

3) In the circuit below, the switch has been open for a long time and is closed at $t = 0$. Let $I_A = 1 \text{ mA}$, $L = 2 \text{ H}$, $R = 1.5 \text{ k}\Omega$, and $C = \frac{1}{6} \mu\text{F}$.



- a) Find the initial conditions $i_L(0^-)$ and $v(0^-)$, and draw the s -domain equivalent circuit for $t \geq 0$.
- b) Find the zero-state and zero-input components of $V(s)$, and then find $v_{zs}(t)$ and $v_{zi}(t)$ for $t \geq 0$.

Marginal Stability & Resonance

4) problem 1

- a) marginal stability part
- b) resonance part
- c) etc

5) problem 2

- a) marginal stability part
- b) resonance part
- c) etc

Fourier Transforms

6) Find the Fourier transform:

$$g(t) = \frac{6}{t^2 + 9}$$

7) Find the output:

$$y(t) = e^{-2t}u(t) * \text{rect}\left(\frac{t}{2}\right)$$